

Bullstreet Academy — TODAY

Session date: Sunday, 4 October 2026 **Repo:** `D:\MY LIFE\SOFTWARE PROJECTS\HAGGAI_TRADING_WEBSITE`
Remote: `https://github.com/OPOKA-ERIC/Bullstreet-Trading-Academy.git` **Live site:** `https://opoka-eric.github.io/Bullstreet-Trading-Academy/` **Supabase project ref:** `bjariigetjvymnvkkes` **Supabase URL:** `https://bjariigetjvymnvkkes.supabase.co` **Remote HEAD at end of session:** `4b4af99`

This document supersedes the earlier `today.pdf` written at 11:05 today. It records the whole day including every failure, because the failures are the useful part: five of the six database errors today were self-inflicted, and the reasons why are worth not repeating.

1. Where things stand right now

Area	State
Frontend JS/HTML/CSS	Working and deployed. All 9 JS files pass <code>node --check</code> .
GitHub Pages routing	Fixed. Relative redirects work from the repo subpath.
Landing page auth buttons	Fixed. Point at the real Supabase auth pages.
Schema migrations	Written, ordered correctly, ASCII-clean — but never successfully applied.
Live Supabase database	Not functional. <code>public.test_questions</code> does not exist.
App end-to-end flow	Untested and presumed broken, because the database is empty.
CLI access to the project	Blocked. Logged in to the wrong Supabase account.

One-line summary: the entire client side is done; the entire database side is not applied; the blocker is that the Supabase CLI is authenticated to an account that does not own this project.

2. The goal

Make this flow work end to end:

signup / login → Impulse Test → diagnosis → prescribed task path → submission → grading → next task unlocks

3. Successes

3.1 Frontend made functional — commit 6635c2a

14 files, +994 / -393. The largest and most valuable commit of the day.

Routing. Root-absolute redirects (`/app/login.html`) 404'd because the site is served from a repository subpath, not a domain root. All internal navigation now uses relative paths.

Access. The landing page "Student Login" and "Create Account" buttons now go to the real Supabase auth pages. The legacy modal login no longer fakes a `localStorage` session — it forwards to `app/login.html`.

Prescription. Added `js/prescription.js` (shared diagnosis metadata + task assignment). The dashboard renders the student's task path grouped by day, with `locked` / `ready` / `in-review` / `passed` states. Completing the Impulse Test auto-seeds the matching task path. Passing a task unlocks the next one.

Tasks. MCQ tasks now render selectable options and are graded against `tasks.correct_answer`. Before this, **every** submission auto-passed. Locked tasks can no longer be submitted by pasting a direct URL, `max_attempts` is enforced, upload failures surface to the user, the retry button genuinely swaps to an alternate task variant, and the verification-unlock notice is populated.

Security. Escaped DB-derived values in the dashboard and grading queue — there was a stored XSS vector via `innerHTML`.

Cleanup. `student.html` and `lesson.html` reduced to redirects into the app.

3.2 Seed data no longer depends on `day_id` — commit ccbec85

The pre-existing live `tasks.day_id` column is `uuid`, but the schema file declared it `integer`. Because `create table if not exists` is a no-op on an existing table, the integer seed values failed. Ordering now comes from a global `sort_order` plus a new `day_label` text column. `day_id` is kept as a nullable passthrough for compatibility and is no longer read by the app.

Also removed `grading.js`'s reliance on an embedded `tasks(day_id)` join to resolve day grouping.

3.3 `ON CONFLICT` targets now exist — commit 6e06a64

`verification_links`, `submission_rubric` and `profiles` also pre-date the schema file, so their declared `PRIMARY KEY` s were never applied. Any `ON CONFLICT` or `supabase-js` `upsert` with `onConflict` against them failed. Each table is deduped by `ctid`, then a unique index is created explicitly. The `verification_links` seed was also switched off `ON CONFLICT` onto the `NOT EXISTS` guard the other seeds use, so a missing constraint can never abort the migration again.

3.4 RLS infinite recursion fixed — commit 96fb02a

This was the important one. The original design was:

```
current_role() -> reads role FROM public.profiles
profiles policy -> calls public.current_role()
```

Evaluating the policy re-entered the policy, so Postgres aborted every authenticated query with `42P17 infinite recursion detected in policy for relation "profiles"`. Signup, dashboard, assessment, task loading and grading were all dead.

The fix moves the authorisation lookup into its own `user_roles` table whose policy is a plain `user_id = auth.uid()`. Nothing ever reads another user's role row, so no cycle can form. `profiles.role` is kept in sync so the client code is unchanged.

While fixing it I also closed a hole I was about to introduce: the `profiles_update_own` policy lets any student `UPDATE` their own profile row, which meant a student could set `role = 'admin'` on themselves. There is now a `BEFORE UPDATE` trigger that reverts role changes unless the caller is already a tutor or admin. It deliberately no-ops when `auth.uid()` is null so that role changes made from the SQL Editor still apply — the first draft of that trigger silently swallowed admin role management, which would have been a worse bug than the one being fixed.

3.5 Function-before-policy ordering fixed — commit `25944cc`

`CREATE POLICY` resolves function references at creation time. The `user_roles` policies were declared *before* `current_role()` existed, so any database without the function already present failed with `42883`. Both `0001` and `0002` were reordered to: create table → create function → create policies.

3.6 Clean-slate reset added — commit `7f3617f`

`supabase/reset.sql` drops the entire `public` schema surface — tables, functions, triggers, policies, storage policies. `auth.users` is deliberately untouched, and `0001` now re-creates profile rows for any login that already exists, so existing logins keep working.

3.7 Read-only diagnostic added — commit `9973ca7`

`supabase/diagnostic.sql` — six catalog queries reporting which functions, tables, RLS flags, policies, column types and row counts actually exist. Intended to stop guessing.

3.8 SQL forced to pure ASCII — commit `4b4af99`

Em-dashes had been silently corrupted into the three-character sequence `U+00E2 U+20AC U+201D` by a PowerShell encoding round-trip. Several landed **inside SQL string literals** (task titles, seed prompts) where the garbage terminates the literal early and the remaining words get parsed as SQL. All SQL is now pure ASCII, and the seeded titles were updated together with their `where t.title = ...` lookups so they still match.

Verified after the fix: no mojibake sequence remains in any tracked text file, and all 9 JS files pass `node --check`.

4. Failures

Six database errors today. Every one is recorded with its actual cause.

#	Error	Root cause	Fixed?
1	42804 column <code>day_id</code> is <code>uuid</code> but expression is <code>integer</code>	Live column type differed from schema file; <code>create table if not exists</code> did nothing	Yes — <code>ccbec85</code>
2	42P10 no unique or exclusion constraint matching <code>ON CONFLICT</code>	Declared PKs never applied to pre-existing tables	Yes — <code>6e06a64</code>
3	42P17 infinite recursion detected in policy for relation <code>profiles</code>	Genuine design flaw in the RLS policy graph	Yes — <code>96fb02a</code>
4	42883 function <code>public.current_role()</code> does not exist	Policies created before the function	Yes — <code>25944cc</code>
5	42601 syntax error at or near <code>as</code>	My diagnostic SQL was malformed. Bad column alias	Yes — diagnostic rewritten
6	42P01 relation <code>a</code> does not exist	Cause never confirmed. Mojibake was one candidate; the error recurred after that fix	No — unresolved
7	42P01 relation <code>public.test_questions</code> does not exist	<code>0001</code> aborted before creating any tables	Consequence of #6

4.1 Error #6 in detail — the one still open

`0001_init.sql` failed with `relation "a" does not exist`, and afterwards `public.test_questions` did not exist either, meaning the script aborted *before* line 57 where that table is created.

Everything before line 57 is only: create `user_roles`, create `current_role()`, grant, enable RLS, three policies, create `profiles`. None of that references an identifier `a`.

What was actually done to investigate:

- Re-read all 549 lines of `0001_init.sql`. No bare `a` in any relation position.
- The only `a` aliases in the file are the three dedupe statements (`delete from public.user_progress a using public.user_progress b ...`), and `DELETE FROM tbl a USING tbl b` is valid PostgreSQL.
- Wrote a proper lexer (`sqllex.js`) that walks the file handling line comments, dollar-quoted blocks and `''`-escaped string literals. Result: **no unclosed literals** in any of `0001_init.sql`, `0002_role_rls_fix.sql`, `reset.sql` or `diagnostic.sql`.
- Ran it across all four SQL files. All clean.

The strongest remaining hypothesis is that **the error was never in the file** — that a 549-line script was damaged in transit during copy-paste into the Supabase SQL Editor. Supporting evidence: the very first diagnostic query pasted back came back with a syntax error in a construct I had written correctly, which proves the paste path is lossy.

That hypothesis is untested, and testing it requires CLI access rather than more guessing.

4.2 The mistake that cost the most time

For five rounds I kept patching migrations against a database whose contents I could not read, inferring the state from whatever error came back. Each error message was treated as a new fact about the schema. This is backwards: the diagnostic should have been run *first*, and the failure was mine — I built a diagnostic, then asked for only its last line rather than all six outputs.

By roughly the fourth round the patching had made things worse, because each fix was applied to a half-applied state and could not be reasoned about. The reset script in §3.6 exists because of that.

5. The blocker for tomorrow

The Supabase CLI is installed and working:

```
supabase --version -> 2.118.0
```

But `supabase projects list` returns only:

Reference ID	Name	Region
<code>jsjhgwficdrgzwbwzkhm</code>	TutorUG	West EU (Ireland)
<code>lgfrlvooobnszfhafzjzkhm</code>	ShopLedger	Central EU (Frankfurt)

`bjariigetjvymnvkkes` — the Bullstreet project the code points at (`js/supabase-client.js:2`) — **is not in that list**. The CLI is authenticated to a different Supabase account than the one that owns this project.

`supabase/.temp/` contains only `cli-latest`, no `project-ref`, so the project is **not linked**. Credentials live in the Windows credential manager and cannot be read from the shell, so the logged-in identity cannot be confirmed programmatically.

A likely reason the first attempt at `supabase login` appeared to succeed while changing nothing: the CLI was already authenticated, so `login` refused and did not switch accounts.

6. Resume here tomorrow

Step 1 — fix the CLI login

Run from the project folder:

```
cd "D:\MY LIFE\SOFTWARE PROJECTS\HAGGAI_TRADING_WEBSITE"
supabase logout
supabase login
supabase projects list
```

`login` prints a URL and waits. Sign in **with the account that owns Bullstreet**. To find that account: open the Supabase dashboard on the Bullstreet project, then click the avatar top-right. If the browser auto-logs into the

wrong account, sign out in that tab or use a private window.

Gate: `supabase projects list` must show the Bullstreet project. If it still shows only TutorUG and ShopLedger, stop and report — the login landed on the wrong account.

Step 2 — link

```
supabase link --project-ref bjariigetjvymnvkkesk
```

Prompts for the database password: Supabase dashboard → Bullstreet project → **Project Settings** → **Database**.

Step 3 — inspect before changing anything

Do this before `db push`, because the live schema is unknown and error #6 is still unexplained:

```
supabase db dump --file supabase/_live_schema.sql --schema public
```

Read it. That single file answers every open question — whether `test_questions` exists, which policies are present, what `tasks.day_id` actually is, and whether the mojibake is present in the live database as well as the repo.

Step 4 — apply

Once the live schema is known, decide between:

- `supabase db push` to apply `0001` then `0002` as versioned migrations, or
- running the SQL directly if the live schema needs a bespoke repair.

Prefer `db push`: it records applied versions, so the same statement never runs twice.

Note that `0002_role_ri_fix.sql` was written as a delta against a database state that no longer exists. On a clean database `0001` alone should be sufficient, and `0002` may be deletable.

Step 5 — ignore `supabase/.temp/`

Currently untracked. Add `supabase/.temp/` to `.gitignore` before committing — it is CLI connection state and should not be in the repo.

Step 6 — verify

```
select 'questions' as t, count(*) from public.test_questions
union all select 'tasks', count(*) from public.tasks
union all select 'rubric_items', count(*) from public.rubric_items
union all select 'task_variants', count(*) from public.task_variants
union all select 'verification_links', count(*) from public.verification_links
union all select 'profiles', count(*) from public.profiles
union all select 'user_roles', count(*) from public.user_roles;
```

Expected: 10 questions, 13 tasks, 4 rubric items, 1 task variant, 1 verification link, and one `profiles` + `user_roles` row per existing login.

Then promote the owner account and confirm the role propagates to both tables:

```
update public.profiles set role = 'tutor'
where id = (select id from auth.users where email = 'academybullstreet@gmail.com');

select u.user_id, u.role as user_roles_role, p.role as profiles_role
from public.user_roles u
join public.profiles p on p.id = u.user_id;
```

Both role columns must read `tutor`. If `user_roles_role` is `tutor` but `profiles_role` is not, the sync trigger is not firing.

Step 7 — test the real flow

- 1. Sign up a brand-new account.
- 2. Confirm the dashboard loads without a recursion error.
- 3. Take the Impulse Test.
- 4. Confirm a diagnosis is saved and the Day 1 task appears as `ready`.
- 5. Submit Day 1.
- 6. Grade it as tutor.
- 7. Confirm Day 2 unlocks.
- 8. Confirm a student cannot self-promote by updating their own profile.

7. Commit log for the day

Time	Commit	Subject
11:05	61c30e1	Add implementation summary PDF (today.pdf)
13:14	6635c2a	Fix GitHub Pages 404s and make the rehab app functional end-to-end
13:34	ccbec85	Fix tasks seed against the live uuid day_id column
13:38	6e06a64	Add missing unique indexes for ON CONFLICT targets
13:46	96fb02a	Fix RLS infinite recursion on profiles (42P17)
13:49	25944cc	Define current_role() before the policies that reference it
13:51	9973ca7	Add read-only catalog diagnostic for the Supabase project
13:58	7f3617f	Add clean-slate reset script and backfill profiles for existing logins
15:58	4b4af99	Strip mojibake from SQL files; it was breaking the parser

All are pushed to `origin/main`. Working tree is clean apart from untracked `supabase/.temp/`.

8. Files touched today

Created

File	Purpose
<code>js/prescription.js</code>	Shared diagnosis metadata + task assignment logic
<code>supabase/migrations/0001_init.sql</code>	Canonical schema, RLS, storage policies, seeds (549 lines)
<code>supabase/migrations/0002_role_qls_fix.sql</code>	Delta repair for the recursion + leftover seeds (225 lines)
<code>supabase/diagnostic.sql</code>	Read-only catalog inspection
<code>supabase/reset.sql</code>	Clean-slate teardown (70 lines)

Modified

`js/app-utils.js` , `js/assessment.js` , `js/auth.js` , `js/dashboard.js` , `js/grading.js` , `js/task.js` , `script.js` , `index.html` , `app/dashboard.html` , `app/task.html` , `student.html` , `lesson.html`

Deliberately not touched

`js/supabase-client.js` — still contains the public anon key. This key is designed to be public and is protected by Row Level Security, which is precisely why the RLS work in §3.4 matters. Once RLS is verified working, confirm the anon key cannot read other users' rows.

9. Known gaps, not attempted today

These were never in scope today and remain unbuilt:

- **Payments.** Flutterwave integration is designed but not implemented. Design notes are in `docs/PAYMENT_AUTH_ARCHITECTURE.md`.
- **Video lessons.** No video hosting or playback.
- **Tutor task-authoring UI.** Tutors can grade but cannot create or edit tasks from the app; that requires direct database access today.
- **Email verification and password reset.** Not configured.
- **`public.days` table.** Exists in the live database with an unknown shape. Nothing reads it. `day_id` was abandoned rather than reverse-engineered. If the day grouping is ever restored, backfill with: `update tasks t set day_id = d.id from days d where d.order_idx = ...`

10. Rules worth keeping

1. **Read the database before changing it.** One dump beats six rounds of inference.
2. **Never paste a large migration by hand.** Split it, or use the CLI. The paste path is lossy.
3. **SQL files stay pure ASCII.** One encoding round-trip cost two hours today.

4. **Define a function before any policy that calls it.** `CREATE POLICY` resolves references eagerly.
5. **Never let a role lookup read the table whose policy calls it.** That is the recursion in §3.4.
6. **Keep migrations idempotent** — `if not exists` , `NOT EXISTS` guards, explicit unique indexes.

11. Session — Tuesday, 6 October 2026

Client-facing changes requested by the account owner. Two items.

Item	Request	Status
Social links	Add Telegram / email / YouTube / Instagram to the site	Done
Registration flow	Register successfully -> land logged in directly, no email confirmation loop	Done (client-side)

11.1 Changes

Social links — added everywhere:

Channel	Link
Telegram	<code>https://t.me/+bqYoAejxzANmOGY8</code>
Email	<code>mailto:academybullstreet@gmail.com</code>
YouTube	<code>https://www.youtube.com/@Bullstreetacademy</code>
Instagram	<code>https://www.instagram.com/__kenny_black?igsh=MWRkeH15Z3ltYXdtnQ%3D&utm_source=qr</code>

- `index.html` — new social row under the footer brand description.
- `app/dashboard.html` — already had the four links in a "Connect" column; fixed the raw `&` in the Instagram URL to `&`.
- `app/login.html` , `app/signup.html` , `app/assessment.html` , `app/task.html` , `app/grading.html` , `app/lesson.html` — the six app pages that had **no footer** now have one, matching the dashboard footer including the "Connect" social links.
- `styles.css` — added `.footer-socials` (wrapped link row, hover in brand green).

Registration flow (`js/auth.js`) — the signup handler now:

1. `signup` + `ensureProfile` as before.
2. If `data.session` exists, straight to `dashboard.html` (project has confirmation off).
3. Otherwise it calls `signInWithPassword` with the same credentials and goes to `dashboard.html` . The success message is now "Account created. Signing you in...".
4. If auto-login still fails, the real reason is shown and the user is sent to `login.html` .

The old message ("Check email if confirmation required") is gone, so a user never sees a dead end after registering.

11.2 Tests run

- All JS files pass `node --check` (including `js/auth.js`, `script.js`).
- Local HTTP smoke test: served the repo and fetched all 8 HTML pages plus `styles.css`, `js/auth.js`, `js/supabase-client.js`, `logo.jpeg` — all returned **200**.
- Confirmed the four social links are present on all 8 HTML pages.
- `app/dashboard.html` Instagram link now HTML-valid (`&`).

11.3 Still true / one caveat for the client

Auto-login only works while Supabase has email confirmation **disabled**. The client hit "email limit exceeded" earlier because Supabase's free tier rate-limits confirmation emails — that symptom is confirmation being on. Fix (dashboard side, cannot be done from code or CLI while the CLI is still linked to the wrong account):

Supabase dashboard -> Authentication -> Providers -> Email -> toggle **off** "Confirm email".

Until that is flipped, a brand-new signup will return `Email not confirmed` in step 3 and the user is redirected to the login page (no dead end, but no straight-through either).

11.4 Committed

Pushed to `origin/main` with the work in §11.1. See the commit log in this repo.

12. Session follow-up — Tuesday, 6 October 2026 (client feedback)

The client tested the site and asked for a hard model change: **no self-registration at all**. Accounts exist only after payment; everyone flows through login.

12.1 What changed

- **Registration removed.** The "Create Account" links are gone from the homepage mobile menu and footer. `app/signup.html` is no longer a registration page — it now redirects to `app/login.html` and states that credentials are issued only after enrollment/payment. `js/auth.js` lost its signup handler and the now-dead `ensureProfile()` helper.
- **Login is the only door.** `app/login.html` now tells users "No self-registration. Your login credentials are issued after you complete enrollment and payment" with an "Enroll now" link back to the pricing section. If a session already exists the page forwards straight to `dashboard.html`.
- **Assessment already required a login** (`assessment.js` calls `requireAuth()` and bounces to `login.html`), so the "Take Assessment" path now behaves exactly as requested: visitor clicks it -> lands on login -> signs in with the credentials issued after payment -> test.

12.2 Test user — created, but blocked by the client-side project setting

The client asked for a fixed test login. I created it on the live Supabase project:

Field	Value
Email	<code>bullstreets@gmail.com</code>

Field	Value
Password	123456789
User ID	55369259-acf7-4d99-aff5-bac96fd2cb1c
Status	Unconfirmed — confirmation_sent_at was set and the signup returned no session

A password grant against that user returned **HTTP 400**. Diagnosis from the API responses: the Supabase project still has **"Confirm email" turned ON**. That single setting is why every signup the client tried "refused" — the user is created but gets no usable session, and `signInWithPassword` answers `Email not confirmed`.

No service-role key exists in the repo, and the Supabase CLI is still on the wrong account (\$5), so this cannot be fixed from code. The client has a one-toggle fix:

Supabase dashboard -> Authentication -> Sign In / Providers -> Email -> **Confirm email: OFF**

After that toggle: delete the unconfirmed `bullstreets@gmail.com` row in **Authentication -> Users** (a re-signup of an existing email returns no new session), then re-run the signup — it will return a session immediately.

12.3 Tests run

- All JS files pass `node --check`.
- Local HTTP smoke test: `index.html`, `app/login.html`, `app/signup.html`, `app/assessment.html`, `app/dashboard.html` all return **200**.
- No `signup.html` / "Create Account" / `signup-form` references remain in any HTML file.

12.4 Still blocked, unchanged from \$5

The live database schema still has no `test_questions`, `tasks`, etc. until the migrations are pushed with the CLI owned by the right account. So after the email toggle is fixed and the test user logs in, the Impulse Test itself will not load questions until `0001_init.sql` is applied. Both remaining blockers share the same root cause: no authenticated access to the Supabase project.

13. Session round 2 — Tuesday, 6 October 2026 (live diagnosis)

Client said "done bruh, do what you can". Live probing of the project found the two remaining blockers and pinned their exact causes.

13.1 Auth: the client disabled the wrong switch

Probe	Result
POST <code>/auth/v1/signup</code> (bullstreets@gmail.com)	HTTP 400

Probe	Result
POST /auth/v1/token? grant_type=password	HTTP 422 email_provider_disabled — "Email logins are disabled"

Between sessions someone toggled the **Email provider itself OFF** in the dashboard. That blocks ALL email logins, not just signups. The earlier session's diagnosis (Confirm email ON) is now compounded by the provider being off. Both are dashboard-only settings.

Dashboard path: Authentication -> Sign In / Providers -> Email -> toggle **ON** the Email provider. Then, so the existing test account can log in without a verification email, run this once in the SQL Editor: `update auth.users set email_confirmed_at = now() where email = 'bullstreets@gmail.com';`

Unlike pasting a 550-line migration, that one line is safe to paste anywhere.

13.2 Database: no tables at all

Checked `public.test_questions`, `public.tasks`, `public.profiles`, `public.user_roles` through the PostgREST endpoint. **Every one returns HTTP 404 PGRST205 "Could not find the table in the schema cache"**. The schema was never applied — the CLI blocker from \$5 is still live (`supabase projects list` still shows only TutorUG and ShopLedger; the Bullstreet project ref `bjariigetjvymnvkesh` is not linked and not visible).

13.3 What was added: `supabase/live_bootstrap.sql`

A single, lossless, idempotent file = `0001_init.sql` + `0002_role_rls_fix.sql` + a post-bootstrap block that:

1. confirms the test account instantly (`email_confirmed_at = now()`), and
2. promotes `bullstreets@gmail.com` to `tutor` in **both** `profiles` and `user_roles` (the guard trigger does not sync `user_roles` when the SQL Editor runs it, so the bootstrap updates that table explicitly).

`0002` was re-verified: on a clean database it is a no-op (every statement is guarded), so running `0001` then `0002` then the helpers is safe. The file is pure ASCII (0 non-ASCII bytes), matching rule 3. Pasting it from the GitHub **raw** view is lossless, unlike typed or copy-pasted-from-chat content.

Applying it creates: 15 tables + trigger + role-guard, RLS policies, the private `submissions` storage bucket + policies, 10 impulse-test questions, 12 rehab tasks, the Day 1 rubric, the Day 1 -> Day 2 verification link, and the Day 1 retry variant, and it backfills `profiles` / `user_roles` rows for every existing login.

13.4 Minimal client action to go live

1. **One dashboard toggle:** Authentication -> Providers -> Email -> **ENABLED** (currently OFF).
2. **One paste:** run `supabase/live_bootstrap.sql` (from GitHub raw view) in the SQL Editor.
3. Then `bullstreets@gmail.com` / `123456789` logs in as tutor with a full, seeded DB, and the student flow (login -> Impulse Test -> prescription -> task -> grading) is testable.

The old "Confirm email" advice from §12.2 is now obsolete — the bootstrap confirms the test user directly in SQL, so the Confirm-email toggle can stay whatever it is.